Vadim HEMZELLEC-DAVIDSON

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Education

École Polytechnique 2025 - Present
Master of Science in Data Science
Relevant coursework: Advanced learning for text and graph data (ALTEGRAD - MVA), Deep Learning, Machine Learning, Optimization, Reinforcement Learning, Representation Learning for Computer Vision and Medical Imaging (MVA), AI for Sound.

Télécom SudParis 2023 - Present
Engineering degree in Applied Mathematics
Relevant coursework: Machine Learning, Continuous Optimization for ML, Stochastic Processes, Bayesian Inference (HMM, Particle Filtering), Applied Statistics.

Lycée Fabert 2021 - 2023
Intensive Preparatory Classes (MPSI–MP*), competitive entrance preparation for top engineering schools.

Research & Academic Projects

Mechanistic Interpretability of the Commitment Decision in Masked Diffusion LMs — Ongoing
- Investigating whether internal states encode token "readiness" better than the output confidence scores current diffusion samplers rely on, via layer-wise linear probes and causal activation patching.
- In collaboration with Transformer Lab (LLaDA-8B, Dream-7B).

OVHai LLM (formerly DragonLLM) — Capstone Project M2 Data Science, 2026
- Created a token-level labeled toxicity dataset of 24K samples of toxic and safe Q&A, leveraging Qwen3-30B models.
- Developed a real-time toxicity and safety classifier to interrupt harmful generations during inference.
- Inspired by Qwen-Guardrails and circuit-breaker architectures for controllable LLM deployment.

Research Paper Studies & Re-implementations, 2025 - 2026
- Re-implemented RLOO with custom baseline and comparison with PPO / GRPO.
- Studied and reproduced core components of CANDI: hybrid continuous/discrete diffusion models on the Code-Parrot dataset.

Hi!ckathon (Hi!Paris) Finalist, 2025
- Finalist and ranked 2nd/68 (tied) in model performance within a team of 6 students.
- Engineered a gated model architecture aggregating a probabilistic router (XGBoost) and a regressor expert to handle zero-inflated target distributions.

Molecular Graph Captioning — ALTEGRAD Competition (MVA), 2025
- Retrieval-based graph machine learning approach to generate natural language captions from molecular graphs.
- Leveraged graph embeddings for molecule–text alignment in a chemistry-informed setting.

Work Experience

Airbus Helicopters, Donauwörth - AI R&D Internship Mar 2026 - Jul 2026
- Designed and implemented an end-to-end pipeline, from raw dataset curation to model training, to fine-tune an AI model for domain-specific XML code generation, within a highly constrained defense environment.
- Curated a highly specialized dataset based on rigorous military specifications to automate functional validation scripts for NH90 helicopter components.
- Implemented the ACE (Agentic Context Engineering) framework to industrialize the solution.

Blockchain AkademIA, Metz - Internship Jun 2025 - Aug 2025
Contributed to the regional democratization of blockchain and AI by supporting the integration of a new solution within the company and leveraging N8N for automation workflows.

Aerow, Paris - Internship Jun 2024 - Jul 2024
Conducted an inventory of potential Research Tax Credit (CIR) eligible topics across all areas of expertise at the company.

Skills and Interests

Prog. Languages	Python, Java, SQL, C, Bash
Frameworks	Pandas, Scikit-learn, Transformers, TRL, vLLM, Matplotlib, Numpy, PyTorch
Soft skills	Leadership (Former VP, Sports Committee), Teamwork (student associations)
Languages	French (native), English (C1), Spanish (B2)
Interests	AI, Sports (Tennis), Music (Piano 10 yrs), Audio/Video production